Helping Math Find Their X: Assessing the Impact of Reasoning Methods, Biases and Translation Systems in Multilingual LLMs

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Abstract

Large Language Models (LLMs) have significantly improved global access to natural language reasoning, text generation, and task completion. While demonstrating strong logical reasoning in English, LLMs are often called upon to support multi-lingual user requests. Therefore, it becomes imperative for the models to show similar strength across languages. This paper presents a comprehensive study in 26 languages on the impact of translation, reasoning strategies, and language resource levels on the mathematical reasoning abilities of multilingual LLMs. By employing both open and closed-source models, we offer novel empirical insights across a range of languages and uniquely into extremely-low resource African languages. Our findings include highlighting critical patterns impacting LLM performance, highlighting areas for improving their multilingual reasoning capabilities beyond their established strengths in English as well as proposing a new baseline reasoning strategy for low-resource languages.

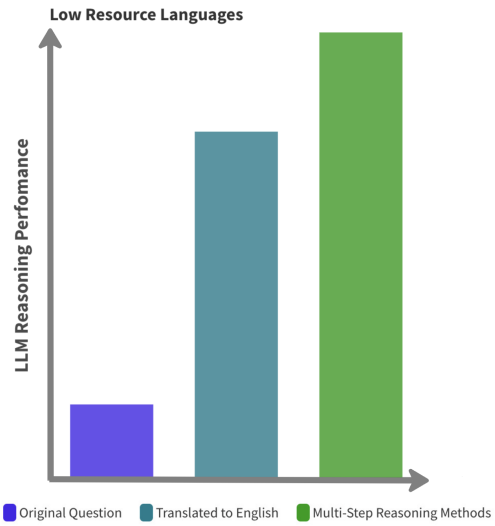


Figure 1: An illustration that highlights our main findings. Although direct translation of the original text to English helps, we find that prompting the models to translate to English as part of their multi-step reasoning results in better performance.

1 Introduction

With the advent of Large Language Models (LLMs), the global access to causal, generative models for task completion and natural language reasoning has grown significantly. (OpenAI et al., 2024; Abdin et al., 2024; Touvron et al., 2023).

Logical reasoning, content understanding and problem solving tasks have become more viable with access to such models in abundance. Furthermore LLMs have shown significant logical reasoning abilities to understand problems, process the input in a stepwise manner and generate text in a multi-lingual settings, while showing exceptional strength in their reasoning abilities in English (Huang et al., 2023; Lai et al., 2023).

However, multi-lingual data used for training these models has a high prevalence of English content. This often leads to sub-par generalization performance when switching from reasoning in English, to other high-resource languages and further lower performance on data from low resource languages and dialects (Lai et al., 2023; Brown et al., 2020; Ahuja et al., 2023).

In this paper, we conduct a comprehensive study on the effects of reasoning strategies and translation models on 26 high, medium, low and extremely-low resource languages. We employ both open and closed source LLMs to understand the effects of these factors across a range of models that have been trained with different objectives and are fine-tuned for instruction following. Our contributions can hence be summarized as follows:

- To the best of our knowledge, this work is the most geographically diverse and expansive multilingual reasoning study to date, encompassing 26 languages across Africa, Americas,

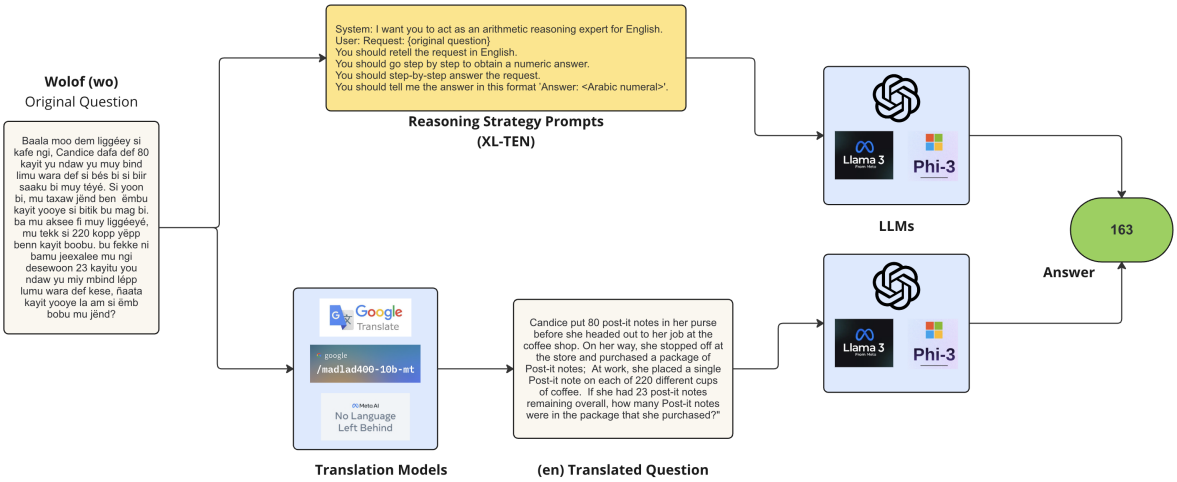


Figure 2: We empirically study the impact and quality of reasoning strategies via prompting and translation systems to assess their ability to use the English-centered bias of LLMs. In this image, a math question in the Wolof (wo) language from the AFRIMGSM dataset is used for representation.

Asia, and Europe.

- As far as we can ascertain, this is also the first empirical work to study the effect on math problem solving within African languages in the context of translation models, inherent biases and prompt-reasoning approaches. These languages represent not only low-resource but in some cases also extremely low-resource scenario as discussed in Section 3.1.

- We also introduce a novel variation of the XLT (Cross-Lingual Thought) prompting strategy called XL-TEN, to improve the quality of multi-lingual reasoning in low-resource settings. Our results show that this variation outperforms XLT on average, and can be considered a new baseline strategy for multilingual reasoning.

- Our work also serves as the first direct exploration into the intensity of translation impact, model and their quality on multi-lingual reasoning as compared to step-by-step reasoning and prompting methods.

2 Related Work

The task of mathematical or logical reasoning has become one of the fundamental metrics for assessing the performance of Large Language Models (LLMs), with performance on various benchmarks such as the GSM8K (Cobbe et al., 2021) and MATH (Hendrycks et al., 2021) becoming a standard accompaniment of the introduction of novel LLMs. The quality of the reasoning abilities of LLMs is often attributed to the corpora they are trained on, which is often English dominated and may have negative impact on the LLM's performance in non-English settings (Zhang et al., 2023).

To assess the scope of this issue, multilingual benchmarks of mathematical were introduced. The MGSM datset (Shi et al., 2022) was created by translating 250 problems from the GSM8K dataset into ten languages, including low-resource ones such as Bengali or Swahili, the MSVAMP dataset was released in (Chen et al., 2023) which contains 1,000 problems translated to the same languages as MGSM[1] and the AfriMGSM dataset (Adelani et al., 2024) which follows the same structure as MGSM but focuses on 16 African languages.

Various approaches for improving the performance of LLMs in non-English settings have been proposed. A commonly used technique is to translate the multilingual input to English and proceed to solve the task as if it was defined in English. The translation can be done either by an external Machine Translation (MT) system (Shi et al., 2022) or by prompting the model to first generate the translation and then solving the problem (Huang et al., 2023; Etxaniz et al., 2023). However, the efficacy of the aforementioned approaches in the context of mathematical reasoning are generally only eval-

[1]With the exception of Telugu (te), which is not included in MSVAMP.

	# of questions	# of questions per language	Languages
MGSM	2,750	250	bn, de, en, es, fr, ja, ru, sw, te, th, zh
MSVAMP	5,000	500	bn, de, en, es, fr, ja, ru, sw, th, zh
AFRIMGSM	4,000	250	am, ee, ha, ig, rw, ln, lg, om, sn, st, sw, tw, wo, xh, yo, zu

Table 1: Summary statistics of the datasets employed in our study. Note that in the case of MSVAMP we sample 500 questions per language even though the dataset does provide 1,000 for each. The rationale behind this decision is elaborated in Section 9.

uated on the MGSM dataset, which may lead to limited generalizability of the obtained results. Furthermore, while (Liu et al., 2024) found that that translation quality has an outsized impact on the final performance, it only compares one open-source MT model with the commercially available Google Translate.

In this study, we build upon these studies and provide a comprehensive evaluation of the said approaches across three distinct datasets encompassing 26 languages and further study the impact of translation quality on downstream performance across five open source MT models whose quality is contrasted with Google Translate.

3 Data

In order to provide a comprehensive assessment of the mathematical reasoning capabilities of various LLMs, we use three distinct datasets across 26 non-English languages. Two of them (MGSM, AFRIMGSM) are translated version of a subset of the GSM8K dataset (Cobbe et al., 2021), a standard benchmarks of mathematical reasoning in English. We introduce the datasets below while the summary statistics can be found in Table 1.

MGSM The first GSM8K-based dataset is the MGSM Benchmark (Shi et al., 2022)[2], with data in 10 translated languages such as Bengali (bn), Chinese (zh), French (fr), German (de), Japanese (ja), Russian (ru), Spanish (es), Swahili (sw), Telugu (te), and Thai (th), along with English (en).

AFRIMGSM The second, AFRIMGSM, is a first-of-its-kind mathematical benchmark in 16 African Languages from diverse parts of the continent, such as Amharic (am), Ewe (ee), Hausa (ha), Igbo (ig), Kinyarwanda (rw), Lingala (ln), Ganda (lg), Oromo (om), Shona (sn), Sotho (st), Swahili (sw), Twi (tw), Wolof (wo), Xhosa (xh), Yoruba (yo) and Zulu (zu), along with English (en).[3] This dataset is part of the African Languages NLU benchmark - IrokoBench (Adelani et al., 2024).

Language	Code	Speak. (M)	CC Size (%)	Cat.
English	en	1,452	45.9693	H
Russian	ru	258	5.9154	H
German	de	134	5.5710	H
Japanese	jp	125	4.9833	H
Spanish	es	548	4.6177	H
French	fr	274	4.2157	H
Chinese	zh	1,118	4.0651	H
Thai	th	60	0.3988	M
Hindi	hi	602	0.1856	M
Bengali	bn	272	0.1078	M
Telugu	te	95	0.0208	L
Amharic	am	60	0.0033	X
Hausa	ha	88	0.0025	X
Kinyarwanda	rw	15	0.0023	X
Zulu	zu	14	0.0022	X
Xhosa	xh	11	0.0019	X
Swahili	sw	71	0.0090	X
Igbo	ib	31	0.0006	X
Oromo	om	45	0.0006	X
Shona	sn	14	0.0006	X
Sotho	st	8	0.0006	X
Lingala	ln	20	0.0002	X
Ganda	lg	6	0.0002	X
Wolof	wo	16	0.0001	X
Yoruba	yo	47	0.0008	X
Twi	tw	17	-	Ø
Ewe	ee	5	-	Ø

Table 2: List of languages, language codes and number of speakers in millions sorted by their proportion in the CommonCrawl corpora and classified as High Resource (H, > 1%), Medium Resource (M, > 0.1%), and Low Resource (L, > 0.01%), Extremely-Low Resource (X, < 0.01%) and Unresourced (Ø, not in the CC list).

MSVAMP The third corpus we study empirically is MSVAMP (Chen et al., 2023), another standard math benchmark in 10 languages that overlap with the MGSM corpus.

3.1 Understanding "low-resource"

In Table 2, we look at statistics of the 26 languages under consideration in this study to provide a perspective on what it means for a language to be

[2] https://github.com/google-research/url-nlp/tree/main/MGSM, https://huggingface.co/datasets/juletxara/MGSM

[3] https://huggingface.co/collections/masakhane/irokobench-665a21b6d4714ed3f81af3b1

called "low-resource," especially when compared to English. To this end, for each of the considered languages, we list their number of speaker as well as the proportion of data in this language in the CommonCrawl corpus [4]. Following (Bang et al., 2023) and (Lai et al., 2023), we also categorize the language based on their resource levels.

As Table 2 shows, while some of the languages can be considered high- or medium-resource, the majority of the considered languages can be classified as either low- or extremely low-resource, with the Twi and Ewe languages not being part of the list of of languages covered by CommonCrawl, despite having 17 and 5 million speakers, respectively. This suggests that the evaluated models will need to cope with languages they were unlikely to encounter during training.

4 Models

We perform empirical analysis with six LLMs, with an equal split between open source and closed-source models (three each). Each of these models have been configured through fine-tuning to follow instructions, or instruction-tuning (Wei et al., 2022)).

GPT Models (OpenAI et al., 2024) We employ three GPT models - GPT-3.5-Turbo, GPT-4 and GPT-4o. These models were chosen to represent the current state-of-the-art in logical reasoning. We utilize Azure OpenAI Service[5] to prompt these models while setting the API version to `2023-12-01-preview`.

Phi3 Models (Abdin et al., 2024) We use Phi3 Mini (3.8B params) and Phi3 Medium (14B) from the Phi3 family of open-source Small Language Models (SLMs) trained on the Phi3 dataset. Despite their size, these models are notable for reporting performance which is comparable and at times also exceeds that of much larger models, particularly on reasoning tasks.

Llama3-70B An open-source 70B parameter instruction-tuned model from the Llama family (Touvron et al., 2023) trained by Meta on more than 15 trillion tokens [6]. We chose this model specifically to represent the open model most likely to be closest to the performance of the commercial models, as evidenced by its performance on the Open LLM Leaderboard[7].

The open models were evaluated using the Ollama framework [8] on a single Nvidia H100 GPU. For all of the models we set the temperature to 0 and ask the completion to contain at most 1024 tokens in order to aid reproducibility of our experiments.

5 Impact of Reasoning Methods

Since all the models in this discussion have been trained or fine-tuned to follow instructions, we work with them in a zero-shot approach of prompting. In this section we outline the different reasoning strategies that we have evaluated with their corresponding prompts in Table 3. The results and analysis follow next, with consolidated results in Table 4.

5.1 Reasoning Strategies

Simple prompt with original language instructions (SIMPLE-ORIG) We ask the models the question in the original language and with a simple instruction also in the same language.

Simple prompt with English instructions (SIMPLE-EN) We ask the question in the original language but with a simple instruction in English.

Original language Chain-of-Thought (COT-ORIG) The question and instruction are in the original language but they are followed by the Chain-of-Thought instruction "Let's think step by step" translated to the original language.

English Chain-of-Thought (COT-EN) (Wei et al., 2023) The question and instruction are in the original language but they are followed by the Chain-of-Thought instruction "Let's think step by step" in English.

Cross-Lingual Thought (XLT) In a step-by-step manner, this state-of-the-art prompting technique (Huang et al., 2023) attempts to retell the multilingual query in English and reason through the given problem. It is based on the premise that even multilingual LLMs have a biased performance gain in English reasoning.

[4] The data reflects the `CC-MAIN-2024-22` crawl and was obtained from https://commoncrawl.github.io/cc-crawl-statistics/plots/languages

[5] https://azure.microsoft.com/en-us/products/ai-services/openai-service

[6] https://github.com/meta-llama/llama3

[7] https://huggingface.co/spaces/open-llm-leaderboard/open_llm_leaderboard

[8] https://ollama.com/

Strategy	Prompt
SIMPLE-EN	System: I want you to act as an arithmetic reasoning expert for the language {original language}. User: {original question} You should format your final answer as 'Answer: <Arabic numeral>'
SIMPLE-ORIG	< SIMPLE-EN translated into Original Language>
COT-EN	System: I want you to act as an arithmetic reasoning expert in English. User: {original question} Let's think step by step. You should format your final answer as 'Answer: <Arabic numeral>'
COT-ORIG	< COT-EN translated into Original Language>
XLT	System: I want you to act as an arithmetic reasoning expert for the language {original language}. User: Request: {original question} You should retell the request in English. , You should go step by step to obtain a numeric answer. You should step-by-step answer the request. You should tell me the answer in this format 'Answer: <Arabic numeral>'
XL-TEN	System: I want you to act as an arithmetic reasoning expert for **English**. User: Request: {original question} You should retell the request in **English**. , You should go step by step to obtain a numeric answer. You should step-by-step answer the request. You should tell me the answer in this format 'Answer: <Arabic numeral>'

Table 3: Zero-shot reasoning strategies and corresponding prompts

Cross-Lingual Thought in (EN)glish (XL-TEN) We also take the XLT approach one step ahead in an attempt to exploit the English-focused bias even further and help improve the multilingual reasoning quality. For this purpose, we introduce the strategy, XL-TEN. While the XLT model creates a strong premise for the task to be solved in English, the impetus is for the model to still be a reasoning expert in the original language. In our exploration, in addition to asking the model to retell the request in English (akin to XLT), we also prompt it to be a reasoning expert in English rather than the original language. Experimental results show that this approach, XL-TEN, is able to exploit the English bias of the multilingual models further and improve multilingual reasoning.

5.2 Reviewing the English Bias

To understand if providing the instruction in English elicits the models' English oriented bias, we compare the EN Reasoning Types in Table 4 to their ORIG counterparts. Comparing SIMPLE-EN to SIMPLE-ORIG; and COT-EN to COT-ORIG, we observe that the EN cases outperform across the three datasets and all six models, with the exception of some statistically close results. This adds weight to the hypothesis that the English dominated instruction following ability of these models is greater than the original languages, even if the questions are asked in the same language across both styles of prompting.

Additionally, upon closer inspection of the results on the AFRIMGSM dataset, it also becomes evident that all English based reasoning methods (including XLT, XL-TEN) outperform the corresponding ORIG variants in a difficult extremely-low resource languages by 5 to 15 percentage points.

5.3 Evaluating XL-TEN: A New Low-Resource Baseline

As the state-of-the-art (Huang et al., 2023) prompting strategy for low-resource settings, XLT is expected to outperform most others. However, according to Table 4, while that is the case for closed source models within MGSM, XLT isn't the best bet for open models or the extremely-low resource settings of AFRIMGSM. On the other hand, for both these cases, XL-TEN outperforms XLT while also being better on average than the original lan-

Dataset	Reasoning Strategy	GPT-3.5-Turbo	GPT-4	GPT-4o	Llama3 70B	Phi3 Medium	Phi3 Mini	AVG
MGSM	SIMPLE-ORIG	39.00	82.36	88.80	55.68	32.72	14.24	52.13
	SIMPLE-EN	52.12	82.88	88.68	**60.16**	42.36	21.92	58.02
	COT-ORIG	50.28	81.48	88.88	57.00	33.84	23.08	55.76
	COT-EN	**58.64**	85.04	88.44	58.72	**43.72**	33.40	**61.33**
	XLT	44.04	**85.48**	**89.28**	56.52	29.92	28.12	55.56
	XL-TEN	52.08	82.72	88.92	59.08	24.60	**35.40**	57.13
MSVAMP	SIMPLE-ORIG	67.67	82.18	88.11	84.80	64.12	55.20	**73.68**
	SIMPLE-EN	70.69	84.27	**88.57**	75.98	56.78	41.62	69.65
	COT-ORIG	66.07	83.00	88.31	**85.92**	**65.83**	**71.80**	76.82
	COT-EN	**71.52**	**85.06**	88.33	74.33	57.84	51.93	71.50
	XLT	66.89	84.64	87.87	74.53	57.24	41.58	68.79
	XL-TEN	68.87	81.80	88.04	74.56	57.84	50.27	70.23
AfriMGSM	SIMPLE-ORIG	4.10	31.85	46.17	7.78	0.53	0.85	15.21
	SIMPLE-EN	8.30	43.48	56.60	**10.67**	3.15	2.60	20.80
	COT-ORIG	5.85	34.85	47.73	8.28	0.55	1.12	16.40
	COT-EN	**11.10**	**48.00**	59.55	8.80	**3.25**	3.27	**22.33**
	XLT	5.97	25.35	58.48	8.71	3.24	2.72	17.41
	XL-TEN	8.17	40.08	**60.12**	8.07	3.20	**3.15**	20.47

Table 4: Average accuracy obtained by applying the respective reasoning strategy across the languages included in the explored datasets. The English subsets of the datasets were not considered in evaluation in order not to bias the results. The best result per dataset and model is boldfaced.

guage prompts.

XL-TEN is also better than XLT on average across the models throughout the three datasets of 26 languages. This suggest that XL-TEN serves as a stronger baseline and state-of-the-art going forward for multilingual reasoning in low-resource languages.

5.4 Performance Cliff in Extremely-Low Resource Settings

From Table 4, it is clear that large closed-source models, especially GPT-4 and GPT-4o that have been trained and fine-tuned on vast multi-lingual resources perform significantly better than open models on the MGSM and MSVAMP datasets (85%+ for GPT-4o vs 55-75% for Llama3-70B).

However, the greater than 50% drop across both closed source and open-source models on the extremely-low resource African languages in AFRIMGSM showcases the limitations of these "multilingual LLMs" when the linguistic diversity becomes wider. When working with low and extremely-low resource languages beyond the most commonly used languages online, these models become prone to high-resource language bias. Based on the results of EN vs ORIG cases, to some extent this drop in performance can improved by providing the instructions in EN over the subpar results obtained in the cases with instructions in the original language (ORIG). Although, even with the better instruction following in English, there is significant scope for improvement for multilingual LLMs to understand the semantics of the original language and follow instructions within their scope, as is evidenced by prior art (Adelani et al., 2024).

6 Impact of Translation

With the LLMs' inherent bias towards English due to data imbalance favoring the language while training and fine-tuning, the expectation from the models is that of better performance in English reasoning. To explore this from a different avenue of translation, we employ the aid the of large-scale state-of-the-art machine translation models. Comparing the performance across them leads to multi-fold insights: understanding the intensity of effects that translation quality can have on multi-lingual reasoning, and perception on whether translation is indeed the answer to all multi-lingual reasoning needs for languages across the world. We first discuss the different translation models we plan to use for experimentation and then follow it with results in Table 5 along with qualitative observations.

6.1 Translation Models

We conduct this experiment with 6 translation models across 3 model families, including one closed-source and 5 open source models.

Dataset	Translation Model	GPT-3.5-Turbo	GPT-4	GPT-4o	AVG
MGSM	T-Google	**73.68**	**86.32**	**84.92**	**81.64**
	T-NLLB-600M	45.93	60.80	59.53	55.42
	T-NLLB-1.3B	34.84	43.93	43.53	40.77
	T-NLLB-3.3B	1.05	0.51	0.69	0.75
	T-MADLAD-3B	63.68	79.44	77.32	73.48
	T-MADLAD-10B	57.44	71.84	70.68	66.65
MSVAMP	T-Google	**74.47**	**83.69**	**86.31**	**81.49**
	T-NLLB-600M	58.78	62.92	66.40	62.70
	T-NLLB-1.3B	49.40	53.50	56.28	53.06
	T-NLLB-3.3B	1.06	0.20	0.60	0.62
	T-MADLAD-3B	63.68	79.44	77.32	73.48
	T-MADLAD-10B	69.10	76.04	78.29	74.48
AFRIMGSM	T-Google	**49.44**	**61.44**	**59.76**	**56.88**
	T-NLLB-600M	19.20	26.43	26.10	23.91
	T-NLLB-1.3B	12.32	15.47	16.60	14.80
	T-NLLB-3.3B	1.20	0.68	0.72	0.87
	T-MADLAD-3B	28.00	35.09	33.42	32.17
	T-MADLAD-10B	35.03	38.91	40.50	38.14

Table 5: Average accuracy obtained by applying the respective translation model on the explored datasets. Note that since the translation models translate into English, the English subsets of the datasets were not considered. The best result per dataset and model is boldfaced.

Google Translate (*T-Google*) Using the Google Translate API [9], the original query is translated into English and used for reasoning.

No Language Left Behind (*T-NLLB*) (Team et al., 2022) We also use the NLLB family of open-source translation models from Meta that has been trained on data from 200 languages. We use three flavors of this model, `nllb-200-distilled-600M` (*T-NLLB-600M*), `nllb-200-distilled-1.3B` (*T-NLLB-1.3B*), and `nllb-200-3.3B` (*T-NLLB-3.3B*).

MADLAD-400-MT (*T-MADLAD*) MADLAD-400 (Kudugunta et al., 2023), or "Multilingual Audited Dataset: Low-resource And Document-level" is a multi-lingual dataset trained on documents in 419 languages by AllenAI. This in-turn was used to train *madlad-400-10B-mt* machine-translation model by Google, which was shown to be qualitatively comparable to much larger models such as the 54 billion parameter model `nllb-54B`, despite being much smaller. In this work we explore two variants of this model: `madlad-400-10B-mt` (*T-MADLAD-10B*) and `madlad-400-3B-mt` (*T-MADLAD-3B*).

For all translation models, we use the CoT-EN prompt, with the input original question translated into English using the machine-translation models:

```
System: I want you to act as an arithmetic reasoning expert in English.
User: {translated EN question} Let's think step by step. You should format your final answer as 'Answer: <Arabic numeral>' "
```

6.2 Analysis

As the results in Table 5 show, the application of Google Translate (*T-Google*) has yielded the best results across all of the considered models and datasets, with the difference from the second-best model when averaged across models being as high as 18 percentage points.

With regards to the open source models, we observe that the MADLAD model family reports results higher than the NLLB model family. This difference is particularly pronounced when it comes to the *T-NLLB-3.3B* which reports performance on an order of magnitude lower than all the other models. Upon closer inspection we find that despite being used in the exact same way as the other two NLLB models, the *T-NLLB-3.3B* generally did not respond with a translation but often times repeated a single word and so on, causing the language model to misunderstand the task and hence respond with a

[9] https://cloud.google.com/translate

wrong (or no) answer.

We note that the comparison with Google Translate is not completely fair, as Google Translate does not support the Wolof (wo) language, and hence its processing was skipped when evaluating the AFRIMGSM dataset. Given the significant difference between Google Translate and all the other open machine translation models, however, the omission of this language does not have a significant impact on the final results or the ranking of translation models. Even if we decided to maximally penalize this model by estimating that it caused all models to output wrong answers in all cases, its average accuracy on AFRIMGSM would decrease from 56.88 to 53.33 percentage points, which would not affect the aforementioned conclusions.

7 Discussion

Impact of tokenization As prior work has found, tokenization can have an outsized impact on reasoning abilities (Singh and Strouse, 2024) and its effect is disproportionally more pronounced in the multilingual setup (Toraman et al., 2022). As part of the introduction of GPT-4o[10], OpenAI has announced that the new model also features a new tokenizer, which is capable of better per-token compression than the previous versions.

Given the extensive nature of our experiments, we proceed to empirically evaluate these claims on five languages from the MSGM benchmark which overlap with the languages mentioned in OpenAI's announcement. We focus on the SIMPLE-ORIG prompting strategy, as its application yield a prompt whose contents are solely in the original language. The results can be seen in Table 6.

As the results in Table 6 show, the GPT-4o model does indeed require a smaller number of input tokens to represent the same input as GPT-4 in general. We further note that the observed ratio (r) for Telugu (4.0x) is even better than the one OpenAI claimed (cr) (3.5x) and that the ratios we observe are generally within 0.1 from the presented ones. In line with contemporary work (Ali et al., 2023; Dagan et al., 2024) we hypothesize that the tokenizer improvements may be one of the reasons GPT-4o's strong multilingual capabilities, especially when compared to GPT-4.

[10] https://openai.com/index/hello-gpt-4o/

Language	cr	GPT-4	GPT-4o	r
Telugu (te)	3.5x	706.58 (181.75)	175.81 (40.73)	4.0x
Russian (ru)	1.7x	198.94 (47.96)	126.63 (28.59)	1.6x
Chinese (zh)	1.4x	163.22 (40.88)	113.97 (27.05)	1.4x
Japanese (ja)	1.4x	205.14 (46.46)	152.62 (35.10)	1.3x
German (de)	1.2x	140.90 (32.69)	123.90 (26.97)	1.1x

Table 6: Average input (prompt) token length for GPT-4 and GPT-4o on various languages from the MGSM benchmark. The numbers in parenthesis represent the standard deviation. The column **cr** represents the "claimed ratio" between the token requirements of the two models as announced by OpenAI at GPT-4o's introduction while **r** represents the ratio we observe in our experiments.

8 Conclusion

In this paper, we perform a detailed exploration of the English-centered bias present multilingual LLMs. We study the impact and quality of reasoning strategies via zero-shot prompting and that of translation models to channel this bias into positive multilingual results in mathematical reasoning tasks. We conduct a thorough empirical analysis across 26 languages from across the globe, varying in their nature from high, medium, low to extremely-low resource; with an added focus on African low-resource languages. Additionally we also propose a novel reasoning strategy, XL-TEN, that outperforms the state-of-the-art multi-lingual reasoning of XLT in low-resource settings, creating a new, stronger baseline. We also empirically analyze the impact of tokenization on reasoning across the languages in the closed-source GPT models. Our findings corroborate the hypothesis of the English centered bias in a range of open and closed source LLMs, and suggest ways to overcome this bias, within constraints, using reasoning & prompting strategies and translation models. Our work also highlights that not all reasoning and translation setups are made equal across the global scale of linguistics and the diversity of languages across the world needs more careful consideration. We will release all the code and data necessary to reproduce our experiments after review, in the hopes of fostering future research in this area.

9 Limitations

Our evaluation is limited to at most 500 samples for each language and each dataset. This is done in order to manage the computational costs and at the same time ensure the resulting dataset is large enough for the analysis to be conducted. Al-

though two of our datasets (MGSM, AFRIMGSM) only feature 250 samples per language, we chose 500 in order to be in line with prior work, such as (Liu et al., 2024). When sub-sampling the original dataset we used a fixed seed to ensure the same questions are chosen per language.

Another challenging aspect of this study is to extract the numeric answers from the model responses. Despite being provided with an answer template in the prompts and setting the most restrictive temperature settings (0), the models tend to generate text in different formats often. For e.g. the answer could be '3', '3 oranges' or '3 oranges in 2 days'; making the extraction pattern tricky to identify. We got around this problem in this study using the help of a multi-step regex to identify the correct numeric value in the responses, however this method might not be fool-proof always. In the follow-up research, we will look into better response extraction methods.

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